YaRN Multilingual Training Report

Full Training Run - SLURM Job 18536300

OpenEuroLLM Long-Context Pipeline | LUMI Supercomputer (CSC) | 2026-05-10

1. Job Summary

Job ID	18536300
Cluster	LUMI (CSC), Finland
Partition	standard-g (AMD Instinct MI250X)
Start time	2026-05-10 08:51 UTC
End time	2026-05-10 17:44 UTC
Wall time	~9 hours
Nodes	32
GPUs	256 (8 per node)
Account	project_462000963

2. Model & Training Configuration

Base model	OpenEuroLLM 9B (LlamaForCausalLM)
Architecture	32 layers, hidden 4096, 32 heads, FFN 14336, vocab 262144
Context extension	YaRN | factor=16.0 | 2048 -> 32768 tokens
Checkpoint loaded from	/flash/project_462000963/jouni/checkpoints/oellm-9b-80-20-TP-2-PP-4
Parallelism	TP=2, PP=4, CP=4, Sequence Parallel, Distributed Optimizer
Sequence length	32768 tokens
Global batch size	128
Micro batch size	1
Total iterations	1000
Total tokens seen	~4.19 billion (128 x 32768 x 1000)
Optimizer	Adam beta1=0.9 beta2=0.95 eps=1e-8
Learning rate	1e-5 peak, 1e-7 min, WSD schedule
Warmup	1/10 of training
Cooldown (WSD)	1/5 of training
Weight decay	0.05
Gradient clipping	1.0
Precision	BF16
Activation recompute	Yes (--recompute-activations)

3. Training Data

Pre-tokenized Megatron bin/idx files from HuggingFace dataset birgermoell/oellm-longctx-tokenized-streamed-all-v2, downloaded and merged on LUMI by download_tokenized.sbatch (job 18504569).

Languages	8: Bulgarian (bg), Czech (cs), Danish (da), Estonian (et), Finnish (fi)
Tiers	16k_plus (>=16384 tokens) | 4_16k (4096-16383) | under4k (<4096)
Merged files	24 (8 languages x 3 tiers)
Total size on disk	87 GB
Estimated tokens	~35 billion
DATA_PATH entries	24 (uniform per-language weighting per tier)
Tier weights	16k_plus: 0.50 total | 4_16k: 0.30 | under4k: 0.20
Per-language weight	0.0625 (16k_plus) | 0.0375 (4_16k) | 0.025 (under4k)
Tokenizer	openeurollm/tokenizer-256k (vocab size 262144)
Data location	/flash/project_462000963/bmoell/data_tokenized_hf_multilingual/

4. Training Loss Curve

999 iterations logged. Table shows every 100th iteration plus first and last. Throughput measured on rank 255 (last pipeline stage).

Iter	LM Loss	Grad Norm	TFLOP/GPU	Notes
2	**12.2211**	**73.530**	**16.1**	*warmup - slow due to dataset index build*
100	7.2543	9.115	40.5	*loss already -5 nats from start*
200	5.3297	2.107	40.6	
300	4.8971	1.962	40.6	
400	4.4716	1.613	40.6	
500	4.0961	1.623	40.5	*midpoint checkpoint saved*
600	3.9516	1.310	40.5	
700	3.8473	0.904	40.5	
800	3.6850	1.098	40.5	
900	3.5562	0.639	40.5	*LR cooldown (WSD) begins*
999	3.6024	0.331	40.5	
1000	**3.6643**	**0.331**	**40.5**	*final checkpoint saved*

Final validation loss: 3.5679 | Validation PPL: 35.44

Final test loss: 3.4302 | Test PPL: 30.88

5. Analysis

The model began training at loss 12.22 - significantly above the random baseline of ln(262144) ~= 12.48. This elevated starting loss is expected: the base checkpoint was pre-trained at 2048-token context, and YaRN's modified rotary position embeddings produce unfamiliar attention patterns at 32K context on the first forward pass.

By iteration 100 the loss had already dropped to 7.25 - a fall of 5 nats in under an hour of training. This rapid early descent reflects the model quickly learning the structure of YaRN-scaled positions. Gradient norms peak at 73 in the first iterations then settle to 1-2 by iter 200, confirming stable convergence.

The final training loss of 3.66 and validation PPL of 35.4 indicate the model has genuinely adapted to multilingual long-context data. The test PPL of 30.9 being lower than validation is consistent with the test split containing slightly longer and cleaner documents concentrated in the 16k_plus tier.

Throughput was rock-solid at 40.5-40.6 TFLOP/s/GPU across all 256 GPUs for the entire 9-hour run (~513 tok/s/GPU). The only slow iteration was iter 2 (16.1 TFLOP/s) due to dataset index building on rank 0 - all subsequent iterations ran at ~32 seconds.

6. Outputs

Checkpoint iter 500	/flash/project_462000963/bmoell/yarn-multilingual/checkpoints/iter_0000500/
Checkpoint iter 1000	/flash/project_462000963/bmoell/yarn-multilingual/checkpoints/iter_0001000/
Tensorboard logs	/flash/project_462000963/bmoell/yarn-multilingual/tensorboard/
SLURM stdout	/scratch/project_462000963/bmoell/yarn-multilingual-18536300.out
SLURM stderr	/scratch/project_462000963/bmoell/yarn-multilingual-18536300.err

7. Next Steps

1. HuggingFace conversion

Run convert_checkpoint.sbatch (job 18602184) to convert iter_0001000 to LlamaForCausalLM format. Manually add rope_scaling to config.json:

{"factor": 16.0, "original_max_position_embeddings": 2048, "type": "yarn"}, "rope_theta": 10000

2. Long-context evaluation

Run RULER, Needle-in-a-Haystack, and LongBench on the converted checkpoint to confirm 32K context capability on downstream tasks.

3. Scale data

Add remaining 27 languages via tokenize_tiers.sbatch (ro, uk, sr, hu, pl, ...) and retrain with a broader multilingual mix.

4. LongRoPE search

Run longrope_search_tokenize.sbatch + longrope_search.sbatch for multilingual RoPE factors; retrain with longrope_multilingual.sbatch if results improve on eval.